ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 3.01: Sequences, Series & Binomial Theorem

Name: _____

Q1. The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

[ ] A: 131
[ ] B: 138
[ ] C: 140
[ ] D: 145

Q2. In an arithmetic sequence, the 3rd term is k + 7 , the 6th term is 3k - 1 , and the 11th term is 6k - 8 , where k is a real constant. What is the first term, a_1 , of this sequence?

[ ] A: 6
[ ] B: 10
[ ] C: 19
[ ] D: 26

Q3. An arithmetic sequence a_n and a geometric sequence b_n of non-zero real numbers share the same first term, so a_1 = b_1 . The sequences also satisfy a_5 = b_2 and a_6 = b_3 , with b_2 neq b_1 . If the maximum value of the partial sum S_N = sum_{k=1}^N a_k of the arithmetic sequence is 153 , what is the sum of the infinite geometric series sum_{n=1}^infty b_n ?

[ ] A: 64
[ ] B: 96
[ ] C: 144
[ ] D: 192

Q4. The bar chart below shows the values of the first four terms of an arithmetic sequence, a_1, a_2, a_3, and a_4, where a_1 = 6, a_2 = 11, a_3 = 16, and a_4 = 21. What is the value of the 8th term, a_8, of this sequence?

[ ] A: 36
[ ] B: 41
[ ] C: 46
[ ] D: 48

Q5. The first three terms of a geometric sequence of positive real numbers are k - 1, k + 2, and 3k, in that order. What is the value of the 6th term of this sequence?

[ ] A: 24
[ ] B: 48
[ ] C: 96
[ ] D: 192

Q6. An infinite geometric sequence of positive real numbers has first term a_1 and common ratio r in 0, 1. Let S_1 = sum_{n=1}^infty a_n, S_2 = sum_{n=1}^infty a_n^2, and S_3 = sum_{n=1}^infty a_n^3. If S_1 = 4 and S_2 = 8, what is the value of S_3?

[ ] A: frac{256}{13}
[ ] B: frac{128}{13}
[ ] C: frac{64}{7}
[ ] D: 16

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Q7. The first four terms of an arithmetic sequence are shown below. 4, 11, 18, 25, dots What is the 31st term of the sequence?

[ ] A: 210
[ ] B: 214
[ ] C: 217
[ ] D: 221

Q8. The second term of a geometric sequence of positive real numbers is k , and the fifth term is k^2 - 20 . If the common ratio of the sequence is 2 , what is the sum of the first 6 terms of the sequence?

[ ] A: 155
[ ] B: 160
[ ] C: 315
[ ] D: 630